

The Schur–Cohn Criterion as a Recursive Pandrosion Reduction

Ivan Besevic

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Abstract

We show that the classical Schur–Cohn stability criterion—a polynomial P has all zeros in the open unit disk $|z| < 1$ if and only if $|P(0)| < |P^*(0)|$ and a degree-reduced polynomial $T(P)$ is also stable—admits a natural formulation as a *recursive Pandrosion reduction*. The Schur transform is:

$$T(P)(z) = \bar{a}_n \cdot Q(0, z) - a_0 \cdot Q^*(0, z),$$

where $Q(0, z) = (P(z) - P(0))/z$ is the Pandrosion quotient at anchor $a = 0$ and $Q^*(0, z) = (P^*(z) - P^*(0))/z$ is its reciprocal analogue. This connects the Schur–Cohn criterion to the Eneström–Kakeya duality (Paper 16) and places both within the broader Pandrosion framework for polynomial zero geometry.

1 The Schur–Cohn criterion

Let $P(z) = a_0 + a_1 z + \cdots + a_n z^n$ be a polynomial of degree n with complex coefficients, and let $P^*(z) = z^n \overline{P(1/\bar{z})} = \bar{a}_n + \bar{a}_{n-1} z + \cdots + \bar{a}_0 z^n$ denote the *reciprocal polynomial*.

Theorem 1.1 (Schur–Cohn). *P has all zeros in $|z| < 1$ if and only if:*

- (i) $|a_0| < |a_n|$; and
- (ii) *The polynomial $T(P)$ of degree $n - 1$, defined by*

$$T(P)(z) := \frac{\bar{a}_n \cdot P(z) - a_0 \cdot P^*(z)}{z}, \tag{1}$$

has all zeros in $|z| < 1$.

The condition is recursive: apply (i) and (ii) to $T(P)$, then to $T^2(P) = T(T(P))$, etc., until a constant is reached.

2 Pandrosion formulation

Definition 2.1 (Pandrosion quotient at anchor 0).

$$Q(0, z) := \frac{P(z) - P(0)}{z} = a_1 + a_2 z + \cdots + a_n z^{n-1}. \tag{2}$$

Definition 2.2 (Reciprocal Pandrosion quotient).

$$Q^*(0, z) := \frac{P^*(z) - P^*(0)}{z} = \bar{a}_{n-1} + \bar{a}_{n-2} z + \cdots + \bar{a}_0 z^{n-1}. \tag{3}$$

Theorem 2.3 (Schur transform as Pandrosion difference). *The Schur transform (1) is:*

$$\boxed{T(P)(z) = \bar{a}_n \cdot Q(0, z) - a_0 \cdot Q^*(0, z).} \tag{4}$$

Proof. Since $P(0) = a_0$ and $P^*(0) = \bar{a}_n$:

$$\begin{aligned}\bar{a}_n P(z) - a_0 P^*(z) &= \bar{a}_n [P(0) + z \cdot Q(0, z)] - a_0 [P^*(0) + z \cdot Q^*(0, z)] \\ &= \bar{a}_n a_0 + z \bar{a}_n Q(0, z) - a_0 \bar{a}_n - z a_0 Q^*(0, z) \\ &= z [\bar{a}_n Q(0, z) - a_0 Q^*(0, z)].\end{aligned}$$

Dividing by z gives (4). □

Remark 2.4. The Schur transform is a *weighted Pandrosion difference*: the leading coefficient $\bar{a}_n$ multiplies the Pandrosion quotient $Q(0, z)$, and the constant a_0 multiplies the reciprocal quotient. The condition $|a_0| < |a_n|$ means the “forward Pandrosion” dominates the “backward Pandrosion”, ensuring contraction toward the origin.

3 The Pandrosion Schur chain

Definition 3.1 (Pandrosion Schur chain). The **Pandrosion Schur chain** of P is the sequence:

$$P = P_0, \quad P_1 = T(P_0), \quad P_2 = T(P_1), \quad \dots, \quad P_n = \text{const.}$$

At each level m :

- $Q_m(0, z) = (P_m(z) - P_m(0))/z$ is the Pandrosion quotient;
- $Q_m^*(0, z) = (P_m^*(z) - P_m^*(0))/z$ is the reciprocal quotient;
- $P_{m+1} = \bar{a}_n^{(m)} Q_m(0, z) - a_0^{(m)} Q_m^*(0, z)$.

P is Schur–Cohn stable iff $|a_0^{(m)}| < |a_n^{(m)}|$ for all $m = 0, 1, \dots, n-1$.

Proposition 3.2 (Chain contraction). *If P is stable, then:*

$$\max_{|z|=1} |P_m(z)| \text{ decreases with } m \tag{5}$$

and the zeros of each P_m are strictly inside the unit disk with radii bounded by

$$\max_k |z_k^{(m)}| \leq \frac{|a_0^{(m)}|}{|a_n^{(m)}|} < 1.$$

4 Connection to Eneström–Kakeya duality

Proposition 4.1 (E-K as a special case of Schur–Cohn). *If P has real positive coefficients $0 < a_0 \leq a_1 \leq \dots \leq a_n$, then:*

- (i) $|a_0| \leq |a_n|$ (the E-K condition implies the first Schur step);
- (ii) The Schur transform simplifies to $T(P) = a_n Q(0, z) - a_0 Q^*(0, z)$ with real coefficients.

Proof. For real coefficients, $\bar{a}_n = a_n$, $\bar{a}_k = a_k$, and $Q^*(0, z) = a_{n-1} + a_{n-2}z + \dots + a_0 z^{n-1}$ is the reversal of $Q(0, z) = a_1 + a_2 z + \dots + a_n z^{n-1}$.

The k -th coefficient of $T(P)$ is:

$$t_k = a_n \cdot a_{k+1} - a_0 \cdot a_{n-1-k}.$$

Under the E-K hypothesis $a_0 \leq a_{k+1}$ and $a_{n-1-k} \leq a_n$, so $t_k \geq 0$ when $a_n a_{k+1} \geq a_0 a_{n-1-k}$, which holds by the ordering. □

Remark 4.2 (Pandrosion hierarchy).

Criterion	Pandrosion object	Paper
E-K (closed disk)	$Q(0, z)$ coefficients increasing	16
E-K duality	$Q(1, z)$ coefficients decreasing	16
Schur–Cohn (open disk)	$T = \bar{a}_n Q(0, z) - a_0 Q^*(0, z)$ stable	19
Gauss–Lucas	$\sum Q(\alpha_k, z) = 0 \Rightarrow$ containment	14

All four criteria use the Pandrosion quotient at specific anchors to localise polynomial zeros.

5 Conclusion

The Schur–Cohn stability criterion is a *recursive Pandrosion reduction*: at each step, remove the constant term via $Q(0, z) = (P(z) - P(0))/z$ (the Pandrosion quotient at anchor 0), form the weighted difference $T = \bar{a}_n Q - a_0 Q^*$ with the reciprocal quotient, and check stability. The recursion terminates after n steps (reaching a constant), and stability requires $|a_0^{(m)}| < |a_n^{(m)}|$ at every level.

This places Schur–Cohn alongside Eneström–Kakeya (Paper 16) as instances of the same Pandrosion principle: *zero containment is encoded in the structure of the quotients* $Q(a, z) = (P(z) - P(a))/(z - a)$.

References

- [1] I. Schur, *Über Potenzreihen, die im Innern des Einheitskreises beschränkt sind*, J. Reine Angew. Math. **147** (1918), 205–232.
- [2] A. Cohn, *Über die Anzahl der Wurzeln einer algebraischen Gleichung in einem Kreise*, Math. Z. **14** (1922), 110–148.
- [3] I. Besevic, *The Eneström–Kakeya theorem via Pandrosion differences and duality*, Preprint (2026).